

* Practical - 6 *

-> Demonstrate Knowledge Base and Query System in prolog.

~~Program~~

```
parent(john, mary).
parent(john, steve).
parent(mary, alice).
parent(mary, bob).
parent(alice, carol).
parent(alice, david).
parent(bob, emily).
parent(bob, frank).
```

```
grandparent(GP, GC) :-
    parent(GP, Parent),
    parent(Parent, GC).
```

```
sibling(X, Y) :-
    parent(P, X),
    parent(P, Y),
    X \= Y.
```

```
brother(X, Y) :-
    male(X),
    sibling(X, Y).
```

```
sister(X, Y) :-
    female(X),
    sibling(X, Y).
```

uncle (Uncle, NieceNephew) :-
brother (Uncle, Parent),
parent (Parent, NieceNephew).

aunt (Aunt, NieceNephew) :-
sister (Aunt, Parent),
parent (Parent, NieceNephew).

male (john).
male (steve).
male (bob).
male (david).
male (frank).

female (mary).
female (alice).
female (emily).
female (carol).



* Output

• ? - uncle (Uncle, carol)

-> Uncle = bob

• ? - grandparent (GP, emily)

-> GP = mary

• ? - aunt (Aunt, david)

-> False

• ? - parent (a, b)

-> False